

Problem

Find the Fourier transform of $f(t) = 2 \cos 2\pi t [u(t) - u(t - 1)]$

Step-by-step solution

Step 1 of 2

Consider the following transfer function.

$$f(t) = 2 \cos 2\pi t [u(t) - u(t - 1)]$$

Calculate the Fourier transform of the function.

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} 2 \cos 2\pi t [u(t) - u(t - 1)] e^{-j\omega t} dt \end{aligned}$$

Changing the limits,

$$\begin{aligned} F(\omega) &= \int_0^1 2 \cos 2\pi t e^{-j\omega t} dt \\ &= 2 \int_0^1 \frac{e^{j2\pi t} + e^{-j2\pi t}}{2} e^{-j\omega t} dt \\ &= \int_0^1 (e^{j2\pi t} + e^{-j2\pi t}) e^{-j\omega t} dt \\ &= \int_0^1 (e^{j2\pi t} e^{-j\omega t} + e^{-j2\pi t} e^{-j\omega t}) dt \\ &= \int_0^1 (e^{-j(\omega-2\pi)t} + e^{-j(\omega+2\pi)t}) dt \\ &= \left[\frac{e^{-j(\omega-2\pi)t}}{-j(\omega-2\pi)} \right]_0^1 + \left[\frac{e^{-j(\omega+2\pi)t}}{-j(\omega+2\pi)} \right]_0^1 \\ &= -\left[\frac{e^{-j(\omega-2\pi)} - e^0}{j(\omega-2\pi)} \right] - \left[\frac{e^{-j(\omega+2\pi)} - e^0}{j(\omega+2\pi)} \right] \\ &= -\left[\frac{e^{-j(\omega-2\pi)} - 1}{j(\omega-2\pi)} \right] - \left[\frac{e^{-j(\omega+2\pi)} - 1}{j(\omega+2\pi)} \right] \end{aligned}$$

Step 2 of 2

Simplify further.

$$\begin{aligned} F(\omega) &= \left(e^{-j(\omega-2\pi)} - 1 \right) \left[\frac{-1}{j(\omega-2\pi)} - \frac{1}{j(\omega+2\pi)} \right] \\ &= \left(e^{-j(\omega-2\pi)} - 1 \right) \left[\frac{-j(\omega+2\pi) - j(\omega-2\pi)}{j(\omega-2\pi)j(\omega+2\pi)} \right] \\ &= \left(e^{-j(\omega-2\pi)} - 1 \right) \left[\frac{j2\omega}{(\omega-2\pi)(\omega+2\pi)} \right] \end{aligned}$$

It is known that,

$$\begin{aligned} e^{j2\pi} &= \cos 2\pi + j \sin 2\pi \\ &= 1 \\ &= e^{-j2\pi} \end{aligned}$$

Therefore,

$$\begin{aligned} F(\omega) &= \left(e^{-j\omega} - 1 \right) \left[\frac{j2\omega}{(\omega-2\pi)(\omega+2\pi)} \right] \\ &= \left(e^{-j\omega} - 1 \right) \left[\frac{j2\omega}{(\omega^2 - 4\pi^2)} \right] \end{aligned}$$

Thus, the Fourier transform of $f(t) = 2 \cos 2\pi[u(t) - u(t-1)]$ is

$$F(\omega) = \left(e^{-j\omega} - 1 \right) \left[\frac{j2\omega}{(\omega^2 - 4\pi^2)} \right].$$

[Comments \(2\)](#)



Anonymous

supposed to be $1j\omega$



Anonymous

(not $j2\omega$)